

BANKABILITY ASSESSMENT

Kano Agro-Industrial Solar-plus-Storage Project

5.0 MWp Solar PV + 4.0 MWh Battery Storage | Commercial & Industrial PPA | Kano State, Nigeria

SAMPLE DELIVERABLE - NOT A LIVE CLIENT ENGAGEMENT

This report has been prepared by CircularEnergies Ltd solely to illustrate the structure, analytical method and standard of its financial modelling and bankability work. The project, project company, sponsor and offtake described are hypothetical. Assumptions are representative of Nigerian commercial and industrial solar market conditions but do not relate to any actual transaction, and no reliance should be placed on the conclusions for any investment purpose.

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1. EXECUTIVE SUMMARY

This assessment evaluates the bankability of a 5.0 MWp solar photovoltaic plant with 4.0 MWh of battery energy storage, supplying an Agro-processing facility in Kano State under a 15-year commercial and industrial power purchase agreement. The assessment concludes that the project is bankable on the terms modelled, with senior debt coverage that withstands the full downside envelope tested. Sponsor returns, however, are materially more sensitive than debt coverage, and the tariff is the binding commercial variable.

Conclusion

The project supports senior debt of USD 3.68 million against a total funding requirement of USD 6.09 million, with a minimum debt service coverage ratio of 1.50x over a fourteen-year amortisation profile. This provides 20 basis points of headroom above the 1.30x covenant even in the most severe combined downside case tested, where minimum coverage falls to 1.37x. On a lender's assessment, the transaction is financeable.

Sponsor economics are adequate but not robust. The base case equity internal rate of return of 19.9% clears an 18% hurdle by a narrow margin, generating equity net present value of approximately USD 221,000. Under a 10% tariff reduction the equity returns falls to 14.9%, below hurdle, while debt coverage remains compliant. The asymmetry is the central finding of this assessment: the capital structure protects the lender well before it protects the sponsor.

Headline metrics

Metric	Result	Benchmark / covenant
Total project cost	USD 6,085,915	USD 1.12 per Wp installed, including storage
Senior debt	USD 3,679,830	66% of base capital cost; 60% of total funding
Sponsor equity	USD 2,406,085	Funds IDC, fees, reserve account and working cash
Minimum DSCR	1.50x	Covenant 1.30x - satisfied with headroom
Average DSCR	1.81x	Typical C&I benchmark 1.35x - 1.60x
LLCR at financial close	2.17x	Comfortable loan life coverage
Project IRR (post-tax)	15.2%	Pre-financing, 20-year horizon
Equity IRR	19.9%	Hurdle 18.0% - cleared by 1.9 points
Equity NPV at hurdle	USD 220,586	Positive but thin relative to project scale

2. PROJECT DESCRIPTION

2.1 Configuration

The project comprises 5.0 MWp of ground-mounted solar photovoltaic capacity paired with a 4.0 MWh battery energy storage system, located adjacent to the offtaker's grain processing facility. Storage is sized to shift midday generation into the evening processing shift and to smooth output rather than to provide extended autonomy, which keeps the storage component proportionate to its revenue contribution.

On a P50 basis the site yields 1,750 kWh per kWp annually, giving gross first-year generation of 8,750 MWh. After 97% availability and 3% combined system and storage round-trip losses, net delivered energy in year one is 8,233 MWh, declining at 0.55% annually through panel degradation.

2.2 Commercial structure

Energy is sold under a 15-year power purchase agreement at USD 145 per MWh, escalating at 2% annually and denominated in US dollars. Contracted annual volume of 9,000 MWh exceeds plant output, meaning the offtaker's load absorbs all generation and the plant is not curtailed by contract volume. A take-or-pay floor set at 85% of contracted volume, or 7,650 MWh, provides downside revenue protection that becomes material in later years as degradation reduces output.

The tariff should be read against the offtaker's current alternative. Diesel-fired self-generation in Nigeria currently costs in the region of USD 380 per MWh, so the PPA represents a saving of approximately 62% on displaced energy cost. This margin is the commercial foundation of the transaction and the principal reason the offtaker's payment incentive is strong: default would return the facility to a materially more expensive energy source.

3. CAPITAL COST AND FUNDING STRUCTURE

Total base capital cost is USD 5,575,500, equivalent to USD 1.12 per Wp installed including storage. This sits within the observable range for Nigerian commercial and industrial solar-plus-storage installations at this scale.

Uses of funds	USD	Sources of funds
Base capital cost	5,575,500	Senior debt
Interest during construction	174,792	USD 3,679,830
Arrangement and upfront fees	55,197	
Debt service reserve account	230,425	Sponsor equity
Minimum operating cash	50,000	USD 2,406,085
Total funding requirement	6,085,915	Total sources 6,085,915

Gearing is set at 66% of base capital cost. Because equity funds interest during construction, arrangement fees, the reserve account and working cash, effective gearing against total funding is 60.5%. Sponsors frequently overlook this distinction when sizing an equity commitment, and the difference here is approximately USD 205,000.

4. FINANCIAL ANALYSIS

4.1 Operating performance

First-year contract revenue is USD 1,193,767 against operating costs of USD 180,365, giving EBITDA of USD 1,013,402 and a margin of 84.9%. The margin is characteristic of solar generation, where costs are overwhelmingly fixed and there is no fuel input. Operating costs escalate at 3.5% annually on a blended naira and dollar basis, against tariff escalation of 2%, so the margin compresses gradually across the operating life.

4.2 Debt sizing and repayment profile

Senior debt is sculpted in proportion to pre-tax operating cash flow, with the repayment profile scaled so that its present value at the 9.5% cost of debt equals the facility amount. This produces a constant pre-tax coverage ratio and repays the facility precisely over the fourteen-year tenor. First-year debt service is USD 460,851.

4.3 Taxation and the pioneer relief cliff

The project is assumed to receive pioneer status for three years. Capital allowances are modelled as deferred rather than forfeited during the relief period, with the pool claimed from the first assessable year at 75% of qualifying cost, being the initial allowance of 50% together with the annual allowance of 25%, and the balance in the following year.

The practical consequence is that the project pays no tax until year six, rather than year four as the three-year relief period alone would suggest. Capital allowances shelter assessable profit for two years beyond the expiry of pioneer relief. From year six, tax of approximately USD 260,000 becomes payable and coverage steps down accordingly.

This treatment is a common source of modelling error. Where capital allowances are assumed to be claimed and exhausted during the relief period, the model shows tax commencing immediately on expiry and understates post-holiday cash flow by a material margin. The difference alters the debt sizing decision, not merely the presentation.

4.4 Coverage profile

Debt service coverage is constant at 2.20x through the first five years while the project is shielded from tax, then steps down to 1.66x in year six when tax commences. Coverage declines gradually thereafter to a minimum of 1.50x in year fourteen as operating cost escalation outpaces tariff escalation and generation degrades.

Period	DSCR	Driver
Years 1 - 5	2.20x	Pioneer relief and capital allowances shelter tax
Year 6	1.66x	Tax commences; allowance pool exhausted
Years 7 - 13	1.65x - 1.53x	Gradual margin compression
Year 14 (minimum)	1.50x	Final year of amortisation

5. SENSITIVITY ANALYSIS

Downside cases are run against a debt service profile fixed at financial close, which is the correct basis for testing covenant compliance once the facility is drawn. All four downside cases remain covenant-compliant.

Scenario	Min DSCR	Equity IRR	Assessment
Base case	1.50x	19.9%	Both tests satisfied
Tariff reduced 10%	1.32x	14.9%	Covenant held; equity below hurdle
Generation at P90 (-7%)	1.50x	17.5%	Take-or-pay floor absorbs shortfall
Operating costs +20%	1.43x	18.2%	Absorbed comfortably
Combined downside	1.37x	14.4%	Covenant held; equity materially impaired

5.1 Interpretation

Two observations follow from the sensitivity results, and both bear on how the transaction should be negotiated rather than merely how it should be reported.

- Tariff is the binding variable. A 10% tariff reduction costs five percentage points of equity return, while a 7% generation shortfall costs only 2.4 points and a 20% operating cost overrun costs 1.7 points. Commercial negotiation effort is therefore better directed at tariff level, escalation and indexation protection than at technical performance guarantees.
- The take-or-pay floor is doing real work. Under the P90 generation case, minimum coverage is unchanged from base at 1.50x because the contractual floor of 7,650 MWh binds before coverage deteriorates. The structure converts a technical risk into a contractual one, which is precisely its purpose, and it materially improves the credit profile.

The asymmetry between debt and equity outcomes is the assessment's central conclusion. In every downside case the lender is repaid on covenant terms; in two of the four the sponsor fails its hurdle. A sponsor reviewing this structure should recognise that the coverage headroom is not equity headroom.

6. RISK ASSESSMENT

Risk	Rating	Assessment and mitigation
Offtaker credit	High	Single-counterparty concentration is the dominant risk. Mitigated by a strong economic incentive to perform, given 62% saving against displaced diesel, but requires audited financials, a bank guarantee or letter of credit covering at least three months of invoicing, and step-in rights.
Tariff and revenue	High	The binding sensitivity. Requires a firm USD-denominated tariff with contractual escalation and no unilateral revision rights.
Currency	High	Revenue and debt are USD-denominated while a portion of operating cost is naira-denominated. The material exposure is the offtaker's capacity to service a dollar-linked obligation from naira earnings, an offtaker credit risk expressed as currency risk.
Construction	Medium	Twelve-month build of proven technology. Mitigated by fixed-price turnkey EPC with liquidated damages and 5% contingency.
Resource and technical	Low	Kano solar resource is well characterised and stable. P90 case absorbed by the take-or-pay floor.
Operating cost	Low	A 20% overrun costs 1.7 points of equity return. Mitigated by fixed-fee O&M.
Regulatory and tax	Medium	Returns depend on pioneer status being granted and on capital allowances being preserved through the relief period. Both should be confirmed in writing before financial close, as loss of either materially alters the tax profile.

7. BANKABILITY CONCLUSION

The project is assessed as bankable on the terms modelled. Senior debt of USD 3.68 million is supported with a minimum coverage ratio of 1.50x and loan life coverage of 2.17x at close, and coverage remains above covenant across the full downside envelope tested. The principal credit consideration is not the project's cash generation, which is stable and predictable, but the single-counterparty concentration on the offtaker.

Sponsor economics warrant closer attention than the headline return suggests. A base case equity return of 19.9% against an 18% hurdle provides limited margin, and two of the four downside cases take the return below hurdle while leaving debt coverage intact.

Conditions precedent recommended

- Executed power purchase agreement with a firm USD-denominated tariff, contractual escalation and no unilateral revision rights.
- Payment security covering not less than three months of invoicing, by letter of credit or bank guarantee.
- Three years of audited financial statements from the offtaker, with a credit assessment.
- Written confirmation of pioneer status and of capital allowance treatment through the relief period.
- Fixed-price turnkey EPC contract with liquidated damages for delay and performance shortfall.
- Independent engineer's review of the resource assessment and system design.
- Land title or long-form lease for not less than the PPA term plus five years.

Levers to strengthen sponsor returns

Three levers would improve equity outcomes without impairing the credit. Extending the debt tenor beyond fourteen years would reduce annual service and improve early distributions. Securing a concessional tranche or partial guarantee would reduce the blended cost of debt. Negotiating a tariff floor with inflation indexation

would directly address the sensitivity that drives the downside cases. Of the three, tariff protection is the most valuable, because it addresses the variable to which returns are most exposed.

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